

We show that composition of morphisms in a category is associative.

Let $\mathcal{C}$ be a category and let $f : a \rightarrow b$, $g : b \rightarrow c$, and $h : c \rightarrow d$ be morphisms in $\mathcal{C}$.

We must show:

$$(g \circ f) \circ h = g \circ (f \circ h).$$

Proof:

By the definition of composition in a category, composition is defined so that for any morphisms whose domains and codomains match appropriately, the result is again a morphism.

Thus, for any object x ,

$$((g \circ f) \circ h)(x) = (g \circ f)(h(x)) = g(f(h(x))).$$

Similarly,

$$(g \circ (f \circ h))(x) = g((f \circ h)(x)) = g(f(h(x))).$$

Since both expressions agree on all inputs x , we conclude that

$$(g \circ f) \circ h = g \circ (f \circ h).$$